LogiDFT - How to Use

From Gaussian log files to Supporting-Information text, 3D structures and a balanced reaction energy profile - worked through with the demo data.

What the tool does. Every structure's final energy is **G = HF (single-point log) + thermal correction to Gibbs free energy (frequency log)**. For relative ΔG and the energy profile, every stage is balanced so it has the **same atoms and the same number of electrons** as the reference stage, by adding or subtracting the energy of reagents / by-products (balancing species).

What you need

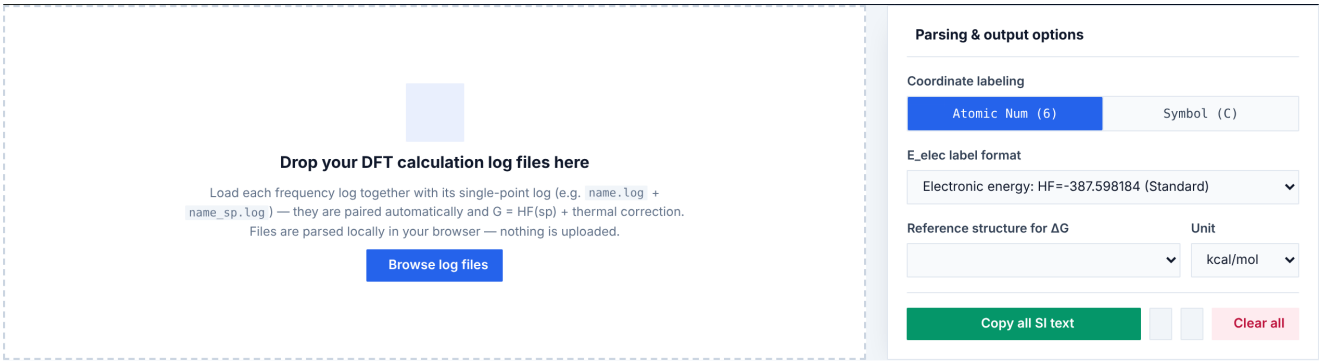
• A **frequency** log (opt+freq or freq) for every structure - gives the thermal corrections and the number of imaginary frequencies.
• A **single-point (sp)** log for the same geometry at the higher level of theory - gives the HF/SCF energy. Name it like the frequency file plus `_sp`: `Int-I.log` + `Int-I_sp.log`. Matching geometry also pairs them.
• The demo files: download `LogiDFT_demo_logs.zip` from the page (empty-state panel), unzip, and follow along. It has 6 structures x (freq + sp) = 12 files.

The demo reaction

Stage	Atoms	Role
Int-I	C5 (30 e-)	Reference stage
TS-1, TS-2	C5	Transition states (1 imaginary freq.)
Int-II	C5	Intermediate
Product	C3 (18 e-)	Lost two carbons - needs balancing
Reagent-B	C2 (12 e-)	Balancing species (the lost C2)

Step 1 - Load the log files

Drop all files on the page, or click **Browse log files** and select them (multi-select is fine). Parsing is done locally in your browser - nothing is uploaded. Load them in the order you want the stages to appear: Int-I, TS-1, Int-II, TS-2, Product, Reagent-B (freq + sp for each). You can reorder later with the arrows in the Table tab.



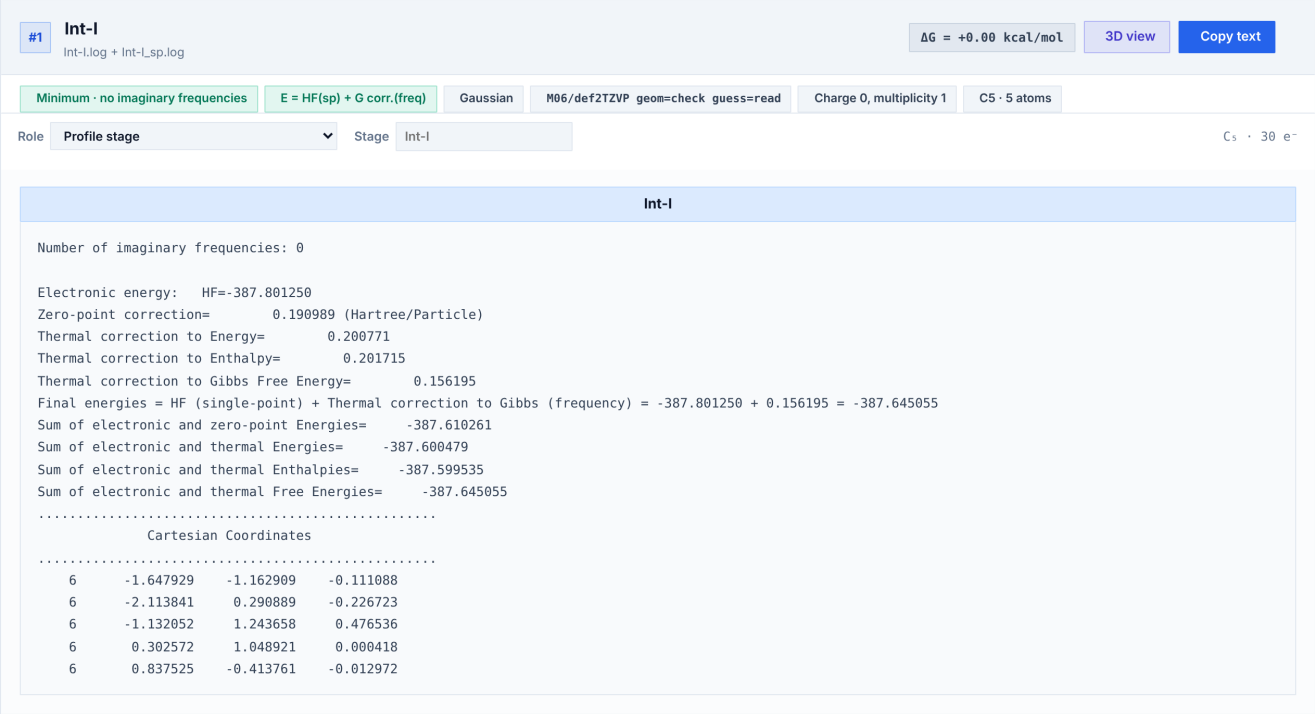
Upload area and options. Coordinate labeling (atomic number or symbol), E_elec label format, the reference stage for ΔG and the energy unit are on the right.

A toast confirms the pairing, e.g. "*Processed 12 log file(s) -> 6 structure(s), 6 with freq + sp paired.*" If a structure shows **No sp log paired** its energy comes from the frequency job - add the matching sp file.

Step 2 - Check each structure

Each structure gets a card. The chips show Minimum / Transition state, the pairing (**E = HF(sp) + G corr.(freq)**), the level of theory, charge/multiplicity and formula. The text block is ready to paste into Supporting Information and ends the energy section with the final energy:

```
Final energies = HF (single-point) + Thermal correction to Gibbs (frequency) = -387.801250 + 0.156195 = -387.645055
```



Card for Int-I. **3D view** opens the structure; **Copy text** copies the SI block. The badge shows the stage's relative ΔG.

Step 3 - Balance atoms and electrons

In the demo, Product is C3 (18 electrons) but the reference Int-I is C5 (30 electrons). The missing C2 (12 electrons) is Reagent-B. On the **Reagent-B** card, change **Role** from *Profile stage* to *Balancing species*:

Before: Reagent-B is treated as a profile stage.

After: Reagent-B is a balancing species (the Stage box disappears).

> **Several species in one stage?** Type the same **Stage** label on each card (e.g. catalyst + substrate -> "Int-I"); their energies are summed. **Which species balance?** Anything that enters or leaves the profile - reagents, ligands, water, by-products. The tool finds the smallest whole-number combination (e.g. + *Reagent-B* or - *2 x Water*) that makes atoms and charge equal to the reference.

Step 4 - Read the balanced stages and the profile

Open the **Table** tab. The upper table lists every species (E(sp), corrections, G). The lower **Balanced stages** table shows, per stage, the composition, the balancing species added or removed, the balanced G and the final ΔG. Choose the reference stage, energy type (G, H, E0, E) and unit at any time.

Energies (Hartree) & relative values

Copy table | CSV

#	Structure	Type	N_{imag}	E_{sp} (HF)	ZPE corr.	H corr.	G corr.	E + ZPE	H	$G = E_{sp} + G_{corr}$	Order
1	Int-I	min	0	-387.801250	0.190989	0.201715	0.156195	-387.610261	-387.599535	-387.645055	
2	TS-1	TS	1	-387.769420	0.188500	0.200044	0.152100	-387.580920	-387.569376	-387.617320	
3	Int-II	min	0	-387.793800	0.190200	0.201244	0.155000	-387.603600	-387.592556	-387.638800	
4	TS-2	TS	1	-387.779100	0.188900	0.200344	0.153400	-387.590200	-387.578756	-387.625700	
5	Product	min	0	-311.784000	0.110000	0.118944	0.080500	-311.674000	-311.665056	-311.703500	
6	Reagent-B	min balancing	0	-75.990000	0.045000	0.048944	0.020500	-75.945000	-75.941056	-75.969500	

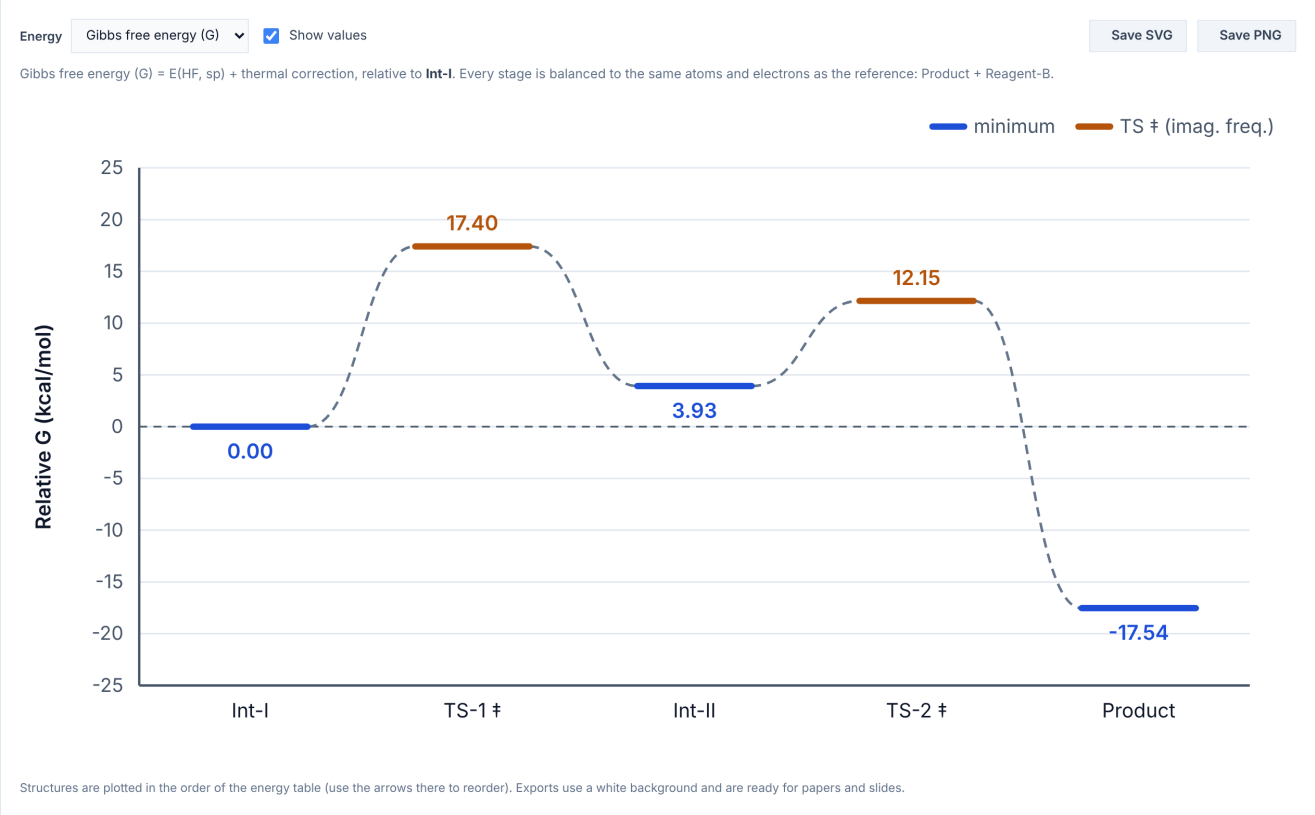
Balanced stages (same atoms & electrons as the reference)

Stage	Species	Composition	Σ G species (Ha)	Balancing species added	Balanced G (Ha)	ΔG (kcal/mol)
Int-I ref	Int-I	$C_5 \cdot 30\ e^-$	-387.645055	—	-387.645055	**+0.00**
TS-1 TS	TS-1	$C_5 \cdot 30\ e^-$	-387.617320	—	-387.617320	**+17.40**
Int-II	Int-II	$C_5 \cdot 30\ e^-$	-387.638800	—	-387.638800	**+3.93**
TS-2 TS	TS-2	$C_5 \cdot 30\ e^-$	-387.625700	—	-387.625700	**+12.15**
Product	Product	$C_3 \cdot 18\ e^-$	-311.703500	+ Reagent-B	-387.673000	**-17.54**

Product: C3 + Reagent-B (C2) = C5, 30 electrons, same as Int-I. ΔG = G(Product) + G(Reagent-B) - G(Int-I).

The energy profile

The **Energy profile** tab plots the balanced stages; transition states are drawn in orange. **Save SVG / Save PNG** export a white-background figure for papers and slides. Stages that cannot be balanced or lack an energy are not plotted and are listed below the chart.



3D structures and exports

• **3D view** on a card opens a ball-and-stick model on a white background. Colours: C gray, H white, N blue, O red, Ni green, B pink. Atoms in the back fade out automatically (depth cue).
• Drag to rotate, scroll to zoom, right-drag to pan. Styles: ball & stick, stick, spacefill, wire. Labels (symbol / number), hide H, measure distance/angle/dihedral (click 2-4 atoms), auto-spin.
• On transition states, **Animate TS mode** shows the imaginary vibration.
• Camera button saves a PNG; **XYZ** copies or downloads coordinates.
• **Copy all SI text** / **Download** produce the SI for all structures; **CSV** or **Copy table** export the energy tables.

Troubleshooting

Message / symptom	What to do
No sp log paired	Add the single-point log (name_sp.log), or make sure both logs have the same geometry.
Stage "cannot be balanced"	Load the missing reagent/by-product (freq + sp) and set it to Balancing species.
"missing energy"	A stage member lacks the sp or the frequency log (thermal correction).
Charge/multiplicity or geometry differ	The freq and sp logs are not for the same species - check the files.
More than 1 imaginary frequency	Higher-order saddle point - re-optimise the structure.
3D view empty	Check the internet connection (3Dmol.js loads from the web) and reload.